

Regression Theory: Comprehensive Guide

Page 1: Introduction

Regression analysis is the workhorse of statistical modeling. It allows researchers to quantify the relationship between a dependent variable (target) and one or more independent variables (predictors). At its core, regression seeks to answer: "How does a change in X affect Y?"

The goal is to find the best-fitting line or curve through a scatter plot of data points.

Page 2: Simple Linear Regression

Simple linear regression focuses on a single independent variable. The model equation is:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Where β_0 is the intercept (the value of Y when X is 0) and β_1 is the slope (the change in Y for a one-unit change in X). The term ε represents the random error, accounting for factors influencing Y that aren't captured by X.

Page 3: The Method of Least Squares (OLS)

How do we find the "best" line? We use Ordinary Least Squares (OLS). OLS minimizes the sum of the squares of the vertical deviations between each data point and the fitted line.

$$SSR = \sum (Y_i - \hat{y}_i)^2$$

By taking partial derivatives of this function with respect to β_0 and β_1 and setting them to zero, we derive the optimal coefficients.

Page 4: Assumptions of OLS

For OLS estimates to be "BLUE" (Best Linear Unbiased Estimators), several assumptions must hold:

- **Linearity:** The relationship is correctly specified as linear.
- **Exogeneity:** The error term has a mean of zero and is uncorrelated with the independent variables.
- **Homoscedasticity:** The variance of the error term is constant across all levels of X.
- **No Autocorrelation:** The errors are independent of each other.

Page 5: Multiple Linear Regression

Real-world problems rarely depend on a single variable. Multiple regression incorporates several predictors:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

This allows us to control for other variables while estimating the specific effect of one predictor.

Page 6: Model Evaluation: R-Squared

How well does our model perform? The Coefficient of Determination (R^2) tells us the proportion of the variance in Y explained by the predictors in the model.

R^2 ranges from 0 to 1. An R^2 of 0.85 means 85% of the variation in the dependent variable is captured by the model.

Page 7: Adjusted R-Squared

R^2 increases automatically as you add more variables, even if they aren't useful. Adjusted R^2 penalizes the addition of unnecessary variables, providing a more honest metric for model complexity.

Page 8: Hypothesis Testing

Are our results statistically significant? We use t-tests to check if individual coefficients (β) are significantly different from zero.

$H_0: \beta_j = 0$ (No effect)

$H_1: \beta_j \neq 0$ (Significant effect)

A p-value less than 0.05 typically indicates that the predictor has a statistically significant relationship with the target variable.

Page 9: Diagnostics and Multicollinearity

Before trusting your model, check for:

- **Multicollinearity:** When independent variables are highly correlated with each other. This makes coefficient estimates unstable.
- **Outliers:** Data points that are far from the regression line and can disproportionately pull the line toward them.

Page 10: Conclusion and Extensions

Regression is a foundational tool, but it is just the beginning. Advanced techniques include Logistic Regression (for classification) and Regularized Regression (Ridge/Lasso) to prevent overfitting. Always remember: "All models are wrong, but some are useful."